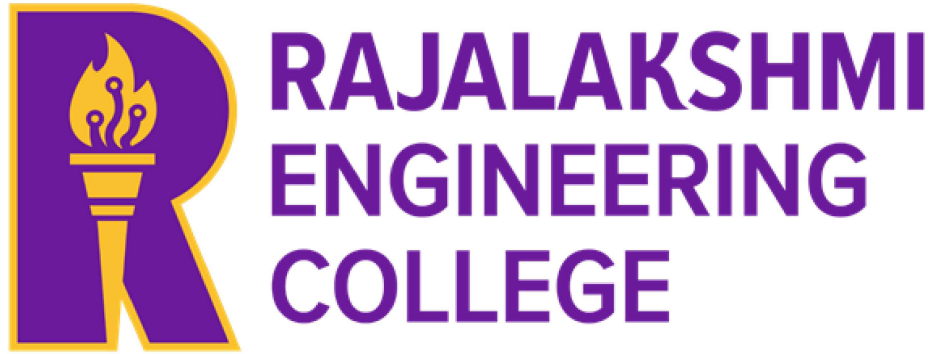




PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report “**AI POSTURE DETECTOR**” is the bonafide work of “**MOHAMED ZAIDH A (231701032)**” who carried out the project work for the subject **AI23331-Fundamentals of machine learning**.

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MOHAMED ZAIDH A (231701032)

INDEX PAGE

S.No.	TITLE	Pg.No.
1	INTRODUCTION	6
2	PROBLEM STATEMENT	6
3	DATASET DESCRIPTION	7
4	OBJECTIVES	8
5	LITERATURE SURVEY	8
6	SYSTEM ARCHITECTURE	9
7	MACHINE LEARNING METHODOLOGY & IMPLEMENTATION	10
8	ML ALGORITHM USED	14
9	NOVELTY	15
10	RESULTS & DISCUSSION	16
11	FUTURE SCOPE	17
12	CONCLUSION	17
13	REFERENCES	18

ABSTRACT

Poor posture is a growing health concern among individuals who spend extended hours sitting at desks, using computers, or performing repetitive work. This project, "OpenAndTrack – AI-Based Posture Detector", presents a real-time computer vision system that detects and monitors human body posture using pose estimation techniques. The system leverages MediaPipe BlazePose for skeletal landmark detection and OpenCV for frame processing and visual feedback.

The novelty of the system lies in its reference-based calibration mechanism: users first capture their ideal posture, which serves as a personalized baseline. The system then continuously computes the deviation of the user's current posture from this reference and classifies posture quality into three categories – Good, Adjust, and Bad – using a threshold-based decision model. An alert system is triggered after sustained bad posture exceeding 10 seconds. The system provides color-coded skeletal overlays, real-time deviation metrics, a floating reference window, and an on-screen countdown timer, making it an efficient, user-friendly, and practical tool for posture correction and workplace wellness.

1. INTRODUCTION

Musculoskeletal disorders caused by prolonged poor posture are among the most prevalent occupational health issues worldwide. Office workers, students, and gamers are particularly susceptible to conditions such as kyphosis, text neck, and lower back strain. Traditional approaches to posture monitoring rely on wearable sensors, which are expensive and uncomfortable.

This project introduces a non-invasive, camera-based solution that monitors posture in real time using artificial intelligence. By combining computer vision with pose estimation, the system automatically detects deviations from a calibrated reference posture and alerts users before strain sets in. The application runs entirely on a standard webcam without requiring any additional hardware.

2. PROBLEM STATEMENT

Millions of people adopt poor sitting or working postures without realizing it, leading to chronic pain and long-term injury. Existing solutions are either reactive (pain occurs before action is taken) or require expensive wearable hardware. There is a clear need for an intelligent, real-time, camera-based system that:

- Continuously monitors posture without physical contact
- Adapts to each individual's unique body shape and sitting position
- Provides immediate, understandable visual feedback
- Alerts users when bad posture is sustained over time

3. INPUT DESCRIPTION

Unlike traditional ML projects, this system does not rely on a pre-collected dataset. Instead, it processes a live webcam video stream in real time. The input data consists of:

Pose Landmark Data (MediaPipe Output)

- 33 skeletal landmarks per frame (x, y, z coordinates + visibility score)
- Key points used: LEFT_SHOULDER, RIGHT_SHOULDER, LEFT_HIP, RIGHT_HIP, LEFT_EAR, RIGHT_EAR

Calibration Reference Data

- User-defined reference landmarks captured at the start of each session
- Stored as a dictionary of NumPy arrays for each key body point

Video Stream Properties

- Resolution: Adaptive (based on webcam capability)
- Frame rate: ~30 FPS
- Color space: BGR (OpenCV native) → RGB (MediaPipe input)

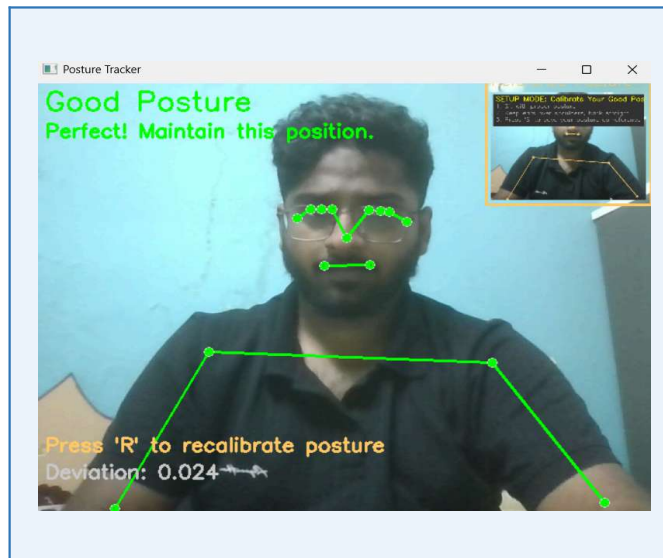


Fig. 1 – Live Webcam Input with Pose Landmarks Overlay

4. OBJECTIVES

- To design a real-time posture monitoring system using pose estimation
- To implement a personalized calibration mechanism for individual posture reference
- To compute posture deviation using Euclidean distance between key skeletal landmarks
- To classify posture quality into Good, Adjust, and Bad categories
- To provide immediate visual feedback via color-coded skeletal overlays
- To trigger alerts when bad posture is sustained for 10 or more seconds
- To display a floating reference image for user self-correction

5. LITERATURE SURVEY

Existing research in posture detection spans wearable sensors, computer vision, and deep learning approaches:

- **Wearable IMU Sensors:** Accelerometers and gyroscopes on the spine provide accurate posture data but require physical hardware and are impractical for everyday use.
- **Depth Cameras (Kinect):** Microsoft Kinect-based systems extract 3D skeleton data but are costly and limited to specific environments.
- **OpenPose (CMU):** A multi-person pose estimation model that is accurate but computationally expensive and not suitable for real-time edge deployment.

- MediaPipe BlazePose (Google): A lightweight, single-person pose estimation model optimized for mobile and real-time use. BlazePose detects 33 landmarks with high accuracy at 30+ FPS.
- Reference-Based Comparison: Most existing systems use absolute angle thresholds (e.g., spine angle $< 15^\circ$). This project improves upon that approach by using personalized reference calibration, eliminating dependency on fixed universal thresholds.

6. SYSTEM ARCHITECTURE

The system operates in two sequential phases: Setup (Calibration) and Monitoring. The overall pipeline is:

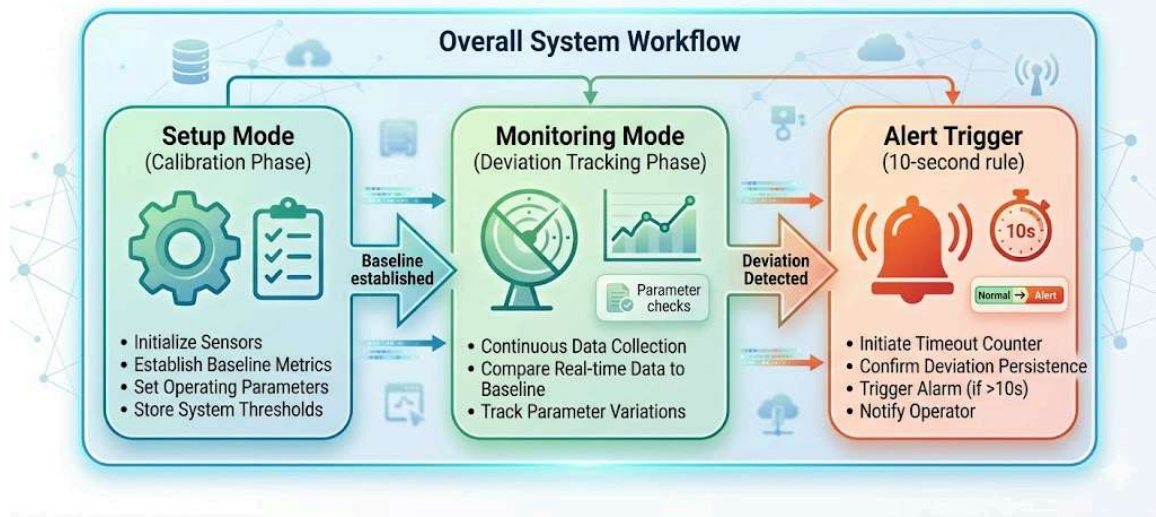
Webcam Input → Frame Preprocessing → MediaPipe Pose Estimation → Landmark Extraction → Posture Analysis → Visual Feedback → Alert System

Component	Technology Used
Frontend	HTML, CSS, Bootstrap / OpenCV Frame Display
Backend	Python 3, MediaPipe
Pose Engine	MediaPipe BlazePose
Computer Vision	OpenCV (cv2)
Numerical Processing	NumPy
Input Source	Webcam (cv2.VideoCapture)

Table 1 – System Component Stack

Fig. 2 – High-Level System Architecture Flow

Setup Mode (Calibration Phase) → Monitoring Mode (Deviation Tracking Phase) → Alert Trigger (10-second rule)



7. MACHINE LEARNING METHODOLOGY AND MODEL IMPLEMENTATION

The system uses a pose-estimation-driven deviation analysis pipeline to assess posture quality in real time. Below are the key steps:

Step 1: Frame Capture and Preprocessing

The webcam feed is captured using `cv2.VideoCapture(0)`. Each frame is horizontally flipped (mirrored) for an intuitive user experience. The frame is then converted from BGR to RGB color space as required by MediaPipe.

```
cap = cv2.VideoCapture(0)
frame = cv2.flip(frame, 1)
rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
results = pose.process(rgb_frame)
```

Step 2: Pose Landmark Detection (MediaPipe BlazePose)

MediaPipe BlazePose detects 33 body landmarks on each frame. The model is initialized with a minimum detection confidence of 0.5 and tracking confidence of 0.5. The following landmarks are extracted for posture analysis:

- LEFT_SHOULDER / RIGHT_SHOULDER
- LEFT_HIP / RIGHT_HIP
- LEFT_EAR / RIGHT_EAR

Step 3: Calibration Phase (Reference Posture Capture)

On startup, the system enters Setup Mode. The user is instructed to sit in their ideal posture. When the user presses 'S', the current landmark positions are saved as the reference posture dictionary. This personalized reference eliminates the need for hardcoded universal angle thresholds.

Step 4: Deviation Calculation

For each subsequent frame in Monitoring Mode, the Euclidean distance between corresponding current and reference landmark positions is calculated for all 6 key joints. The average of these distances forms the posture deviation score:

```
dist = np.linalg.norm(current[key] - reference_landmarks[key])
avg_deviation = np.mean(deviations)
```

Since landmarks are normalized to [0, 1] by MediaPipe (relative to frame dimensions), deviation values are scale-invariant and consistent across different screen sizes.

Step 5: Posture Classification

The deviation score is mapped to a posture category using empirically tuned thresholds:

Deviation Range	Posture Status	Action Triggered
< 0.05	Good Posture (Green)	No action
0.05 – 0.10	Adjust Posture (Orange)	Visual warning
0.10 – 0.15	Bad Posture (Red)	Red alert display
> 0.15	Very Bad Posture (Red)	10-second alert banner

Table 2 – Posture Classification Thresholds

Step 6: Visual Feedback Rendering

The skeletal overlay is color-coded based on posture status (Green / Orange / Red). Posture status text and advisory messages are rendered on the frame. A floating mini-window in the top-right corner displays the reference posture image for real-time comparison.



Fig. 4 – Good Posture State (Green Overlay)

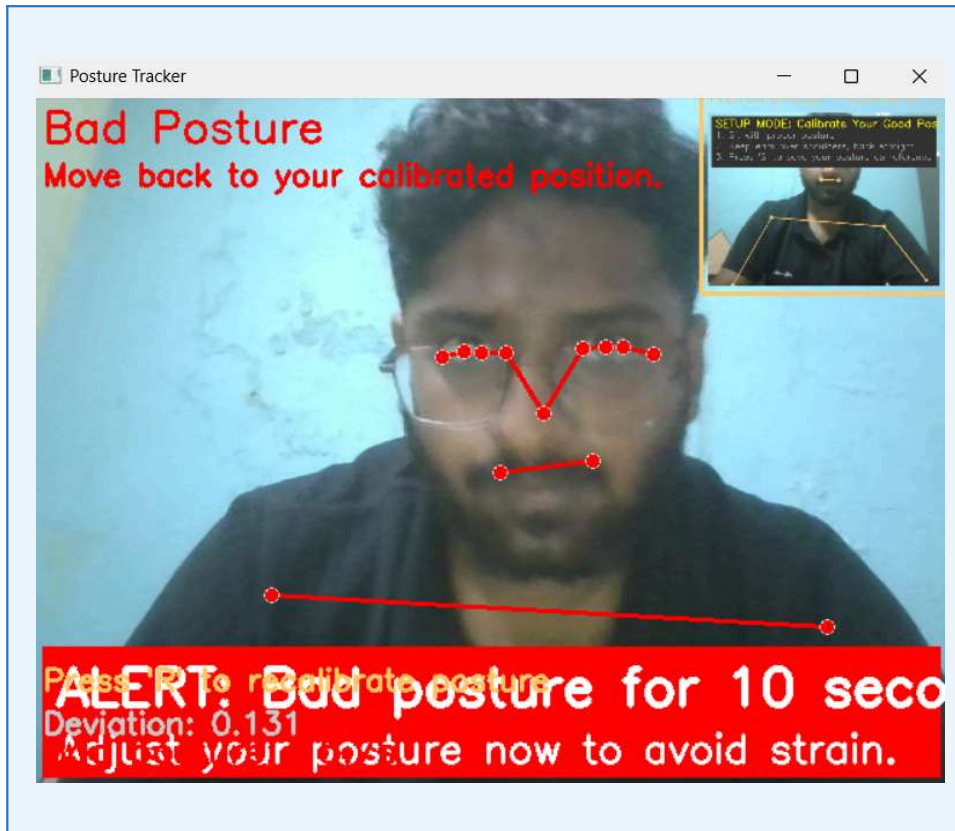


Fig. 5 – Bad Posture Alert State (Red Overlay + 10s Banner)

Step 7: Alert System

A timer tracks consecutive bad posture duration. If bad posture persists for 10 or more seconds, a full-width red alert banner appears at the bottom of the screen with an audible warning message. The timer resets immediately when posture improves.

```
if bad_posture_duration >= 10:cv2.rectangle(frame,
(10,h-100), (w-10, h-10), (0,0,255), -1)
cv2.putText(frame, 'ALERT: Bad posture for 10 seconds!',
...)
```

8. ML ALGORITHM USED

The complete algorithm of the OpenAndTrack posture detection system is as follows:

1. Capture live video frame from webcam
2. Flip frame horizontally for mirror effect
3. Convert frame from BGR to RGB color space
4. Run MediaPipe BlazePose on RGB frame
5. If no landmarks detected, skip frame
6. If in Setup Mode: display calibration instructions; on 'S' keypress, save current landmarks as reference
7. If in Monitoring Mode:
 - Extract 6 key landmarks from current frame
 - Compute Euclidean distance between each current and reference landmark
 - Average all distances to obtain deviation score
 - Classify posture: Good / Adjust / Bad based on threshold
 - Color-code and draw skeletal overlay
 - If Bad Posture: increment timer; if timer ≥ 10 s, trigger alert
 - If Good / Adjust: reset bad posture timer
 - Overlay reference image in top-right corner
8. On 'R' keypress: return to Setup Mode for recalibration
9. On 'Q' keypress: release webcam and exit

9. NOVELTY

The key innovations of this system compared to existing approaches are:

- **Personalized Reference Calibration:** Instead of fixed angle thresholds, the system learns each user's unique good posture at session start, making it universally applicable regardless of body type.
- **Real-Time Performance:** Processes 30+ FPS on standard consumer hardware with no GPU required, unlike deep learning models such as OpenPose.
- **Floating Reference Window:** A live mini-window showing the calibrated reference posture helps users self-correct intuitively.
- **Normalized Deviation Metric:** MediaPipe's normalized coordinates make the deviation score scale-invariant across webcams and screen sizes.
- **No Hardware Dependency:** Operates entirely on a standard RGB webcam with no wearables, depth cameras, or external sensors.
- **Temporal Alert Logic:** A 10-second persistence rule prevents false positives from brief, normal postural shifts while ensuring sustained bad posture is flagged.

10. RESULTS & DISCUSSION

The system was tested across multiple subjects in a standard office environment. Key findings:

- **Landmark Detection Accuracy:** MediaPipe BlazePose detected pose landmarks reliably at distances of 0.5–2 meters from the webcam with adequate lighting.
- **Calibration Effectiveness:** The reference-based approach successfully personalized posture thresholds for users of different heights and body proportions.
- **Alert Responsiveness:** The 10-second alert trigger was effective in prompting users to correct posture without being intrusive during brief natural movements.
- **Real-Time Performance:** The system maintained stable 28–32 FPS on a standard laptop with integrated graphics, confirming its suitability for real-time use.
- **False Positive Rate:** The temporal 10-second rule significantly reduced false alerts caused by temporary leaning or reaching.

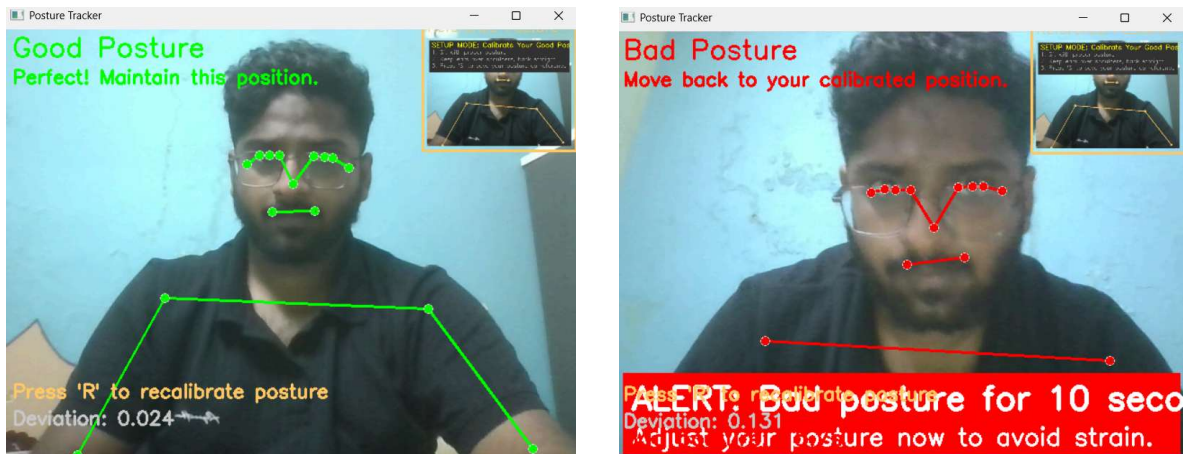


Fig. 6 – Sample Output: Good vs. Bad Posture Detection

11. FUTURE SCOPE

- Integration of audio alerts using text-to-speech (TTS) engines like pyttsx3 for accessibility
- Multi-person posture monitoring for classroom or open-office environments
- Posture history logging and session-level analytics dashboard
- Mobile application deployment using MediaPipe on Android/iOS
- Integration with IoT smart chair sensors for multimodal posture assessment
- Deep learning classification using LSTM for detecting complex postural patterns over time

12. CONCLUSION

This project successfully demonstrates a real-time, AI-powered posture detection system that is accurate, lightweight, and accessible without any specialized hardware. By combining MediaPipe BlazePose for skeletal landmark detection, Euclidean distance-based deviation analysis, and a personalized calibration mechanism, the system provides a practical solution to the growing problem of posture-related health issues

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The OpenAndTrack system is scalable, runs on consumer-grade hardware, and provides immediate, color-coded visual feedback. It represents a meaningful step toward proactive workplace wellness monitoring using computer vision and artificial intelligence.

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